Research Internship Report

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Abstract

This report details my summer internship at Carnegie Mellon University (CMU) from June 2, 2025, to August 22, 2025, under the supervision of Professor Ismaila Dabo. As a Graduate Summer Intern, I contributed to research on equivariant neural networks for atomistic simulations, focusing on mathematical derivations underpinning models such as DimeNet, NequIP, and Allegro. My responsibilities included deriving radial and spherical Bessel basis functions from the Schrödinger equation, computing Clebsch–Gordan coefficients and spherical harmonic products, exploring equivariance in message-passing neural networks for scalar and vector features, and analyzing transformations in spherical coordinates.

A central question was how products of spherical harmonics relate to angular dependencies such as $\cos\theta$: through derivations, I showed that scalar tensor products yield Legendre polynomials $P_\ell(\cos\theta)$, so that for $\ell = 1$ the contraction is proportional to $\cos\theta$, and for $\ell = 2$ it gives a quadratic function of $\cos\theta$. This clarified how bond-angle information is encoded in equivariant representations. The internship bridged my background in software engineering and AI with materials science, advancing my understanding of symmetry-aware neural networks for atomistic dynamics.

1 Introduction

This report provides a comprehensive overview of my internship at Carnegie Mellon University Africa (CMU Africa) as a Graduate Summer Intern in research. The internship aimed to support ongoing projects in equivariant neural networks under Professor Ismaila Dabo, contributing to advancements in atomistic dynamics simulations. CMU Africa, located in Kigali, Rwanda, is a hub for innovative engineering research. The purpose was to apply my academic background in software engineering and my MSIT studies to research problems at the intersection of artificial intelligence and materials science. Objectives included performing detailed mathematical derivations, analyzing key concepts like spherical harmonics and tensor products, and investigating how angular correlations are represented in neural networks for atomistic systems.

2 About the Organization

Carnegie Mellon University Africa (CMU Africa), part of Carnegie Mellon University's College of Engineering, was established in 2011 in Kigali, Rwanda, to address Africa's engineering talent gap. Its mission is to provide world-class education and research opportunities, emphasizing innovation, entrepreneurship, and societal impact. CMU Africa engages in graduate programs such as Information Technology, Electrical and Computer Engineering, and Engineering Artificial Intelligence. It operates as a mid-sized institution with around 200 students and 50 faculty/staff, structured around academic departments and research centers. In the higher education and research sector, CMU Africa focuses on AI, cybersecurity, and recently, materials science. Relevant history includes partnerships with African governments and industries, fostering projects in smart grid technologies, AI for healthcare, and now symmetry-aware AI for scientific computing.

3 Internship Scope and Duration

The internship spanned from June 2, 2025, to August 22, 2025, encompassing 12 weeks of full-time engagement. I was placed in Professor Ismaila Dabo's research group, with a focus on theoretical derivations to support equivariant neural network architectures. My work fit into the group's broader goals of developing scalable AI models for atomistic simulations, including models such as DimeNet for molecular interactions, NequIP for efficient equivariant message-passing, and Allegro for large-scale atomistic dynamics.

4 Roles and Responsibilities

My assigned tasks centered on mathematical derivations and analysis:

- Conducted a literature review on equivariant neural networks, summarizing key papers on local equivariant representations for atomistic dynamics.
- Derived radial and spherical Bessel basis functions starting from the time-independent Schrödinger equation, simplifying to the Helmholtz equation, and applying boundary conditions to obtain functions such as

$$\tilde{e}_{RBF,n}(d) = \sqrt{\frac{2}{c}} \cdot \frac{\sin\left(\frac{n\pi}{c} d\right)}{d}.$$

- Explored equivariance derivations for scalar and vector features in message-passing neural networks, using Clebsch–Gordan coefficients to prove rotational equivariance under SO(3).
- Computed products of spherical harmonics for $\ell_1 = \ell_2 = 1$, using Clebsch–Gordan expansions to show how they decompose into higher-order harmonics.
- Derived transformations from Cartesian to spherical coordinates, including the gradient operator and scale factors.
- Analyzed the Allegro model, deriving how scalar tensor products encode angular correlations through Legendre polynomials.
- Investigated the central question of whether products of spherical harmonics can be expressed as $\cos^\ell \theta$: I showed that the direct product does not equal $\cos^\ell \theta$, but the scalar tensor product yields Legendre polynomials $P_\ell(\cos\theta)$, which for $\ell = 1$ reduces to the dot product $\cos\theta$ and for $\ell = 2$ to a quadratic in $\cos\theta$.

Challenges included handling Clebsch–Gordan coefficients, Wigner 3j symbols, and the addition theorem, which required revisiting group theory concepts I had not studied before.

5 Learning and Experience

Through this internship, I acquired in-depth knowledge of spherical harmonics, tensor products, and representation theory as applied to AI models. I learned to derive basis functions from quantum mechanics principles, prove equivariance using group-theoretic tools, and interpret mathematical structures in terms of physical quantities such as bond angles. This experience highlighted the gap between my software engineering background and the physics-heavy requirements of this field. More exposure to physics and materials science coursework would have eased the process, but the challenge strengthened my ability to self-learn and to connect concepts across disciplines.

6 Achievements and Contributions

Key achievements included producing detailed derivation documents clarifying:

- Radial and angular basis functions for DimeNet.
- Equivariance proofs and message-passing structures for NequIP.
- Tensor product derivations in Allegro, showing explicitly that

  $$(V_{ij} \otimes E_i)_{00} \ \propto \ P_\ell(\cos\theta_{jik}),$$

  with $\ell = 1$ giving $\cos\theta$ and $\ell = 2$ giving a quadratic in $\cos\theta$.

These results advanced the group's understanding of how angular correlations are encoded in local equivariant representations.

7 Challenges and Solutions

The most difficult aspects were grasping the Clebsch–Gordan formalism, interpreting Wigner 3j symbols, and understanding the addition theorem. At times, I reached the limits of my prior understanding, but I overcame these obstacles by breaking derivations into smaller steps, cross-verifying with symbolic computation, and seeking guidance from professor Dabo. The process clarified the role of the Addition Theorem in encoding bond-angle information, which is central to models like Allegro.

8 Insights into the Field

The field of equivariant neural networks is rapidly evolving, aiming for scalable and physically consistent architectures for atomistic simulations. Models such as DimeNet, NequIP, and Allegro show different strategies for combining local geometric features with equivariant constraints. Industry practice emphasizes computational efficiency, while academic research stresses mathematical rigor. My work underscored the importance of balancing these approaches and highlighted how symmetry-aware models can impact applications like materials design and drug discovery.

9 Recommendations and Conclusion

For CMU Africa, I recommend expanding interdisciplinary opportunities by offering more coursework in physics and materials science to complement AI training. This would better prepare students like me to bridge software engineering with domain-specific research in physical sciences. For future interns, I suggest a focus on both mathematical derivations and practical model-building, ensuring a balance of theory and implementation.

In summary, this internship provided a pivotal learning experience. I advanced my understanding of equivariant neural networks, clarified how tensor products of spherical harmonics yield angular dependencies through Legendre polynomials, and contributed derivations relevant to models like DimeNet, NequIP, and Allegro. The work not only deepened my technical skills but also highlighted the importance of interdisciplinary knowledge in tackling frontier research questions in AI for materials science.